Interactions between Dataset Characteristics and Regularization in CNN Image Classification

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Abstract

Regularization is essential to increasing the accuracy of convolutional neural networks and regularization methods are frequently employed in the task of image classification. Our project aims to determine the effect of regularization methods Cutout, RandomErasing, Dropout, Max Dropout, and Label Smoothing on two datasets of varying spatial features, number of categories, as well as degrees of categorical distinction. To do this we employed two CNN architectures: ResNet and AlexNet. The datasets we used were CIFAR-10 (low spatial features, multiclass classification, high categorical distinction), and CelebA (high spatial features, binary classification, low categorical distinction). Through methodical trials of each combination of regularization methods, CNN, and dataset, we identified RandomErasing and Label Smoothing as the best set of regularization methods for CIFAR-10 on ResNet CNN, with test accuracy of 87.22%, and RandomErasing alone as the best for CelebA with test accuracy of 97.46%. For AlexNet CNN, we identified Cutout as the best regularization methods for CIFAR-10, with test accuracy of 81.64%, and Cutout with RandomErasing as the best for CelebA with test accuracy of 97.86%. Our code can be found **here**.

1. Introduction

Our research project aims to investigate the efficacy of employing various regularization methods in conjunction with Convolutional Neural Network (CNN) architectures. The primary objective is to enhance the accuracy of image classification across two distinct datasets characterized by varying levels of spatial features and categorical distinction. Through systematic experimentation and analysis, we seek to identify optimal combinations of regularization techniques and CNN structures that contribute to improved performance in diverse image classification scenarios.

To form our methods and hypothesis, we reviewed "Avoiding Overfitting: A Survey on Regularization Methods for Convolutional Neural Networks" (Santos & Papa, 2018). This paper provides an overview of regularization methods and outlined the results of regularization methods on their own and in various combinations as found in relevant experiments. Existing literature covers a broad spectrum of regularization methods, however those included in the survey were ran only on CIFAR-10, CIFAR-100, ImageNet, SVHN, and differences in performance across datasets were not considered when evaluating regularizer performance.

Our research project delves into unraveling the intricacies of image content and classification categories under the influence of diverse regularization methods and CNN algorithms. Through this exploration, we aim to gain a nuanced understanding of how these factors interact and shape the outcomes in image processing. Our collective effort is to distinguish this research by focusing on the unique interplay between regularization techniques, CNN algorithms, and image content. By shedding light on these dynamics, we anticipate contributing valuable insights to refine and optimize CNN models, advancing the realm of computer vision and image classification.

2. Methods

In the design of our methods, we had three main considerations. First was the datasets to use. We wanted to find datasets with varying characteristics (level of categorical distinction, image complexity, and number of spatial features) as to clearly show the interaction between these characteristics and our regularization methods. Following from this, naturally, was to determine which regularization methods to include. We wanted to include regularization methods in each of the three main areas: data augmentation, internal structure changes, and label augmentation. Lastly we wanted to run our experiments on multiple CNN architectures to account for model (non-data) related differences.

2.1. Datasets

One of the main choices we had to make in the design of our project was was dataset characteristics we wanted to compare as well as what datasets would best these characteristics. After considering a number fo datasets we ended up using two popular ecisiting datasets: CIFAR-10 and CelebA.

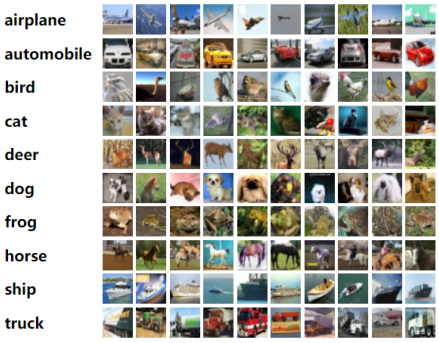

Figure 1. CIFAR-10 Dataset

CIFAR-10 is one of the most popular datasets used in multi-class image classification. It is a subset of the Tiny Images dataset with 60000 images for each of 10 classes. The classes in this dataset are highly distinct including various classes of vehicles and animals. The images are 32x32 and have relatively few spatial features.

Figure 2. CelebA Dataset

CelebA is a human face image dataset containing 202,599 images of 10,177 celebrities. The dataset has 40 (non-mutually exclusive) binary labels. We chose one of these labels, male vs female, to focus on. The classification done here is much less distinct. While airplanes and deer are highly differentiated in visual attributes, labels of male and female can be much more ambiguous. Additionally, the images in this dataset are size 178×218 and thus have a relatively higher number of spatial features.

2.2. CNN Architectures

While not central to our primary hypothesis, the choice of CNN architectures was crucial due to their potential impact on our results. We employed two architectures: ResNet(He et al., 2016) and AlexNet(Krizhevsky et al., 2012). ResNet supports deeper network structures with its skip connections, mitigating vanishing gradient issues and reducing overfitting risks. In contrast, AlexNet, chosen for its simpler structure, offered a balance between training efficiency and accuracy maintenance. This architecture facilitated a more efficient training process without significant accuracy losses, optimizing the trade-off between training duration and performance. Additionally, the shallow network design of AlexNet simplified the analysis of training enhancements, contributing to overall efficiency and effectiveness.
AlexNet is used in 2 configurations for an in-depth exploration:

1. The network with all the regularizations accurately that included dropout and Batch Normalization.
2. The network with a bare-bone CNN inspired by the AlexNet architecture.

2.3. Regularization Methods

Much like datasets, the regularization methods selected would directly shape the conclusions we are able to draw from our experiments. Our main goal in the selection of regularization methods was to cover all three main categories of regularization (data augmentation, internal structure changes, and label augmentation) as well as to include methods that are currently popular for use in Image classification. By achieving these two goals, we can achieve the most representative and applicable results. The regularization methods we employed are (by category):

- Data Augmentation: Cutout, RandomErasing
- Internal Structure Changes: Dropout, MaxDropout
- Label Augmentation: Label Smoothing

We were interested in the effects of Cutout and RandomErasing as they are directly related to the complexity and spatial features of the images in the dataset. For internal structure changes, Dropout and MaxDropout are some of the most popular in CNN literature and Dropout has been one of the most important regularization methods for all types of neural networks(Santos & Papa, 2018). Thus the results will be applicable to many existing models and architectures. Lastly, Label Smoothing is the basis of almost all label augmentation methods. The conclusions we draw about Label Smoothing thus will have implications on label augmentation as a whole.

2.4. Experiments

Our experimental procedure involved training our convolutional neural networks (CNNs) using various regularization methods and combinations thereof, followed by evaluating the performance on the test set.
For AlexNet, we systematically applied each regularization method individually, as well as in combination—specifically,

Cutout with Random Erasing. These combinations were tested both in the presence and absence of dropout and batch normalization layers.
Similarly, in ResNet, each regularization method was applied separately, in pairs from different categories, and in various combinations of one method from each of the 3 categories.

3. Results

We obtained experiment results varying by dataset, CNN, and additional configurations. We will discuss results below by dataset.

3.1. CIFAR-10 Results with ResNet

We ran the CIFAR-10 dataset on our ResNet CNN with depth of 44.

Table 1. CIFAR-10 with ResNet Depth 44 - % Accuracy

DATA AUG	INTERNAL	LABEL AUG	TEST
NONE	NONE	NONE	84.28%
CUTOUT	NONE	NONE	84.27%
ERASING	NONE	NONE	84.25%
NONE	DROPOUT	NONE	**86.81%**
NONE	NONE	SMOOTHING	**84.43%**
CUTOUT	DROPOUT	NONE	**86.27%**
CUTOUT	NONE	SMOOTHING	84.09%
ERASING	DROPOUT	NONE	**86.31%**
ERASING	NONE	SMOOTHING	**87.22%**
NONE	DROPOUT	SMOOTHING	**86.54%**
NONE	MAXDROP	SMOOTHING	**85.80%**
CUTOUT	DROPOUT	SMOOTHING	**87.04%**
CUTOUT	MAXDROP	SMOOTHING	**85.33%**
ERASING	MAXDROP	SMOOTHING	82.14%

With this configuration, we saw the best results from experiments employing DropOut as one of its regularization methods. These results tended to be around 86% to 87%. CutOut and RandomErasing did not increase the test accuracy when employed on their own, but did pretty well when combined with Dropout.Surprisingly, the top test accuracy came from using RandomErasing and Label Smoothing. The lowest test accuracy came from using RandomErasing, MaxDropout, and Label Smoothing. The effect of Label Smoothing is unclear and varies depending on the combinations of other methods employed. On its own it increased test accuracy by just 0.15%.

3.2. CelebA Results with ResNet

For CelebA with ResNet, we ran experiments with a model depth of 20.

Immediately, the baseline accuracy with no regularization is much higher than that for CIFAR-10, even with a decreased model depth. However the effects of regularization are worse across the board. The only combinations that improved test accuracy were RandomErasing on its own

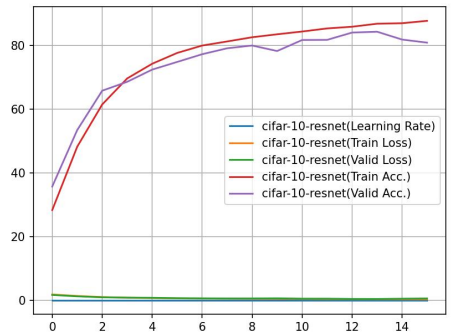

Figure 3. CIFAR-10 with ResNet Depth 44, Baseline no Regularization

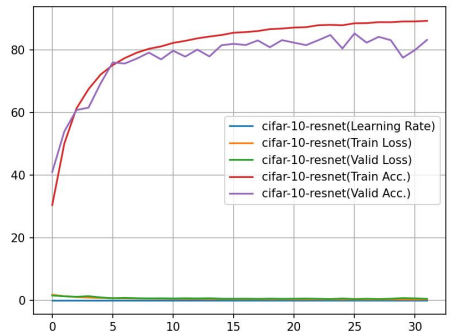

Figure 4. CIFAR-10 with ResNet Depth 44, RandomErasing and Smoothing

Table 2. CelebA with ResNet % Accuracy

DATA AUG	INTERNAL	LABEL AUG	TEST
NONE	NONE	NONE	97.23%
CUTOUT	NONE	NONE	97.14%
ERASING	NONE	NONE	**97.46%**
NONE	DROPOUT	NONE	97.06%
NONE	NONE	SMOOTHING	97.07%
CUTOUT	DROPOUT	NONE	97.21%
CUTOUT	NONE	SMOOTHING	97.23%
ERASING	DROPOUT	NONE	97.12%
ERASING	NONE	SMOOTHING	**97.27%**
NONE	DROPOUT	SMOOTHING	97.19%
NONE	MAXDROP	SMOOTHING	96.44%
CUTOUT	MAXDROP	SMOOTHING	96.81%
ERASING	MAXDROP	SMOOTHING	96.72%

and RandomErasing with Label Smoothing. Experiments containing MaxDropout yielded the lowest accuracy. The best results were with just RandomErasing and the worst was with MaxDropout and Label Smoothing.

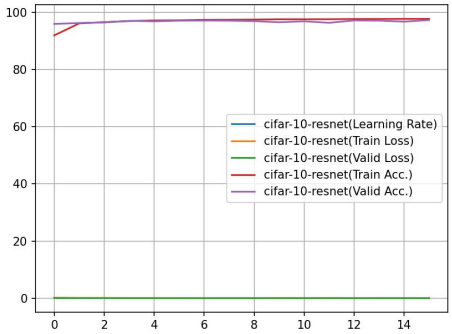

Figure 5. CelebA with ResNet, Baseline no Regularization

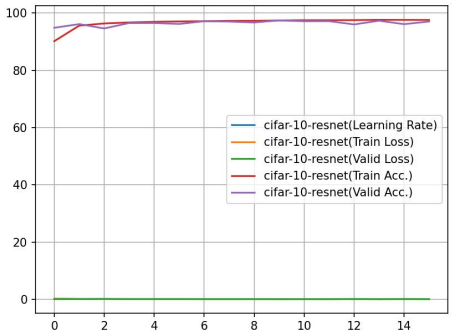

Figure 6. CelebA with ResNet Depth 56, RandomErasing

3.3. CIFAR-100 Results with ResNet

Additionally, for only our ResNet CNN, we ran data augmentation and label augmentation experiements on CIFAR-100, a subset of the same Tiny Images Dataset from which CIFAR-10 was selected but with 10 times as many categories and 10 times fewer images per category.

Table 3. CIFAR100 with ResNet % Accuracy

DATA AUG	INTERNAL	LABEL AUG	TEST
NONE	NONE	NONE	60.97%
CUTOUT	NONE	NONE	**62.56%**
ERASING	NONE	NONE	**61.04%**
NONE	NONE	SMOOTHING	60.54%

Overall accuracy was much lower on this dataset which is expected given the smaller volume of training data per class. Both data augmentation methods increased test accuracy but Label Smoothing lowered test accuracy.

3.4. CIFAR-10 Results with AlexNet

For this section, we have conducted 2 experiments, Experiment 1: AlexNet on CelebA with regularizations and Experiment 2: AlexNet on CelebA without regularizations.

Through this we can observe that the Cutout has consistently performed well on training in CIFAR. The worst performing augmentation is Cutout + Random Erasing through both the experiments.

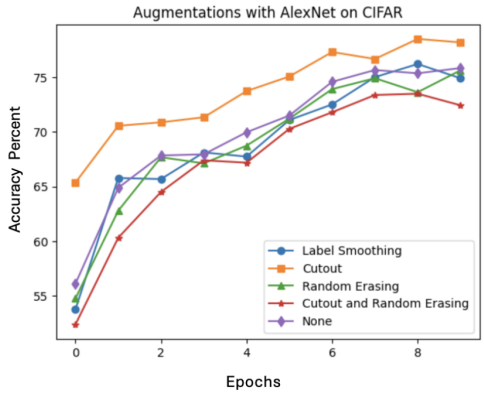

Figure 7. AlexNet with Regularizations on CIFAR

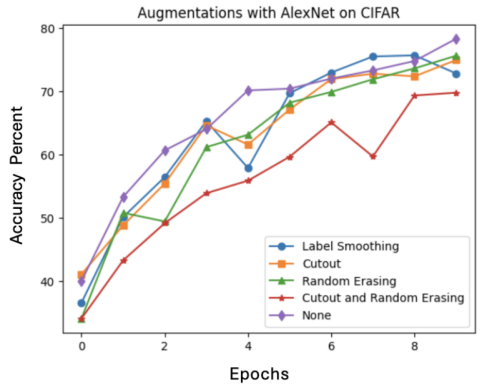

Figure 8. AlexNet without Regularizations on CIFAR

The blue line representing Pure AlexNet consistently achieves higher accuracy scores across different augmentations like without anything, cutout and random erasing, random erasing, cutout, and label smoothing, while the orange line for AlexNet without Dropout BatchNorm lags

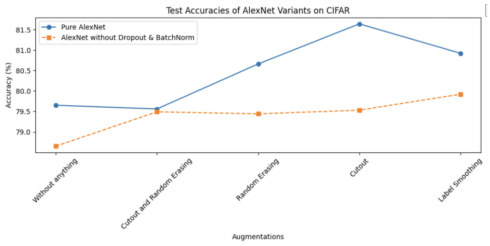

Figure 9. AlexNet Final Test Results CIFAR

behind in performance. This suggests that the Dropout and BatchNorm components play an important role in improving the accuracy of the AlexNet architecture on this particular image classification task. We can also see that the Cutout augmentation has consistently performed better in all cases

Table 4. CIFAR-10 with AlexNet % Accuracy

DATA AUG	LABEL AUG	TRAIN	TEST
NONE	NONE	75.85%	79.65%
CUTOUT	NONE	78.20%	**81.64%**
RANDOM ERASING	NONE	75.64%	**80.66%**
BOTH	NONE	72.44%	79.56%
NONE	SMOOTHING	74.92%	**80.92%**

Table 5. CIFAR-10 with AlexNet without Dropout and BatchNorm2d% Accuracy

DATA AUG	LABEL AUG	TRAIN	TEST
NONE	NONE	78.32%	78.65%
CUTOUT	NONE	74.96%	**79.53%**
RANDOM ERASING	NONE	75.71%	**79.44%**
BOTH	NONE	69.80%	**79.49%**
NONE	SMOOTHING	72.84%	**79.92%**

3.5. CelebA Results with AlexNet

For this section, we have conducted 2 experiments, Experiment 1: AlexNet on CelebA with regularizations and Experiment 2: AlexNet on CelebA without regularizations.

Through this we can observe that the Cutout has consistently performed well on training in CelebA. The worst performing augmentation is Random Erasing through both the experiments.

Based on the graph shown in the image, the conclusion is that the pure AlexNet variant performs significantly better on the CIFAR test dataset compared to the AlexNet without Dropout & BatchNorm variant. The blue line representing Pure AlexNet consistently achieves higher accuracy scores across different augmentations like without anything, cutout and random erasing, random erasing, cutout, and label

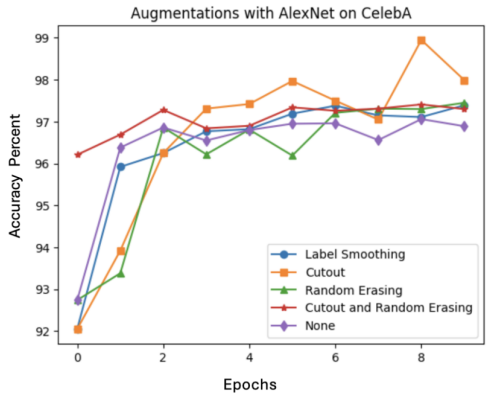

Figure 10. AlexNet without Regularizations on CelebA

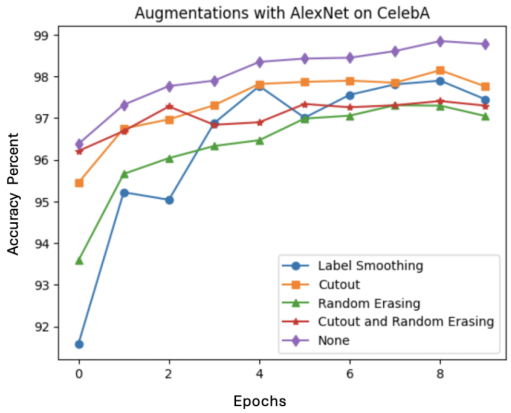

Figure 11. AlexNet with Regularizations on CelebA

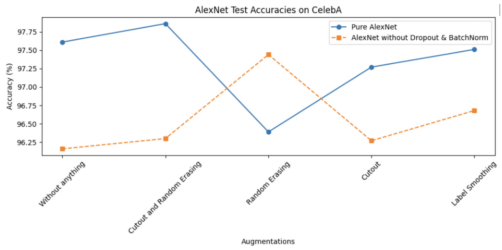

Figure 12. AlexNet Final Test Results CelebA

smoothing, while the orange line for AlexNet without Dropout & BatchNorm lags behind in performance. This suggests that the Dropout and BatchNorm components play an important role in improving the accuracy of the AlexNet architecture on this particular image classification task. We can also see that the Cutout augmentation has consistently performed better in all cases

Table 6. CelebA with AlexNet % Accuracy

DATA AUG	LABEL AUG	TRAIN	TEST
NONE	NONE	98.78%	97.61%
CUTOUT	NONE	97.76%	97.27%
RANDOM ERASING	NONE	97.05%	**96.39%**
BOTH	NONE	97.30%	**97.86%**
NONE	SMOOTHING	97.76%	97.51%

Table 7. CelebA with AlexNet without Dropout and BatchNorm2d % Accuracy

DATA AUG	LABEL AUG	TRAIN	TEST
NONE	NONE	96.89%	96.16%
CUTOUT	NONE	97.99%	**96.27%**
RANDOM ERASING	NONE	97.45%	**97.44%**
BOTH	NONE	97.30%	**96.30%**
NONE	SMOOTHING	97.39%	**96.68%**

4. Process

One of the principal challenges in this study was the construction of the convolutional neural network (CNN) and the selection of an appropriate architecture. Initially, the network was modeled after the architecture of VGG-16. Although VGG-16, being a very deep network, yielded high training accuracies, it presented significant drawbacks in terms of training duration—approximately 2-3 hours per epoch on the CIFAR-10 dataset. Consequently, alternative architectures were explored, leading to the selection of AlexNet. With its shallower network compared to VGG-16, AlexNet delivered respectable results while substantially reducing the training time. Therefore, the decision was made to proceed with AlexNet's architecture.

Another challenge in this study was the data processing in the model. Because of the change from Cifar10 to CelebA, which have different class numbers and image resolution, we have to change our dataloader and modify the model's structure.

Because of the time limitation, we have to abandon experiments on Handwritten Symbols Dataset, which is a math symbol dataset with simple spatial features.

We additionally ran CIFAR-10 with Cutout, RandomErasing, and Label Smoothing individually on our ResNet CNN with a model depth of 56.

With the increased model depth, all 3 experiments showed increased test accuracy, though the improvements weren't as significant as some of those seen with the previous model depth. As this didn't add significant value to our existing experiments with model depth 44, we elected not to include this in our results section.

Table 8. CIFAR-10 with ResNet Depth 56 - % Accuracy

DATA AUG	INTERNAL	LABEL AUG	TEST
NONE	NONE	NONE	88.34%
CUTOUT	NONE	NONE	**88.48%**
ERASING	NONE	NONE	**88.86%**
NONE	NONE	SMOOTHING	**88.37%**

5. Contributions

The CelebA dataset was converted from a multi-label dataset to a binary classification dataset for male or female identification. This transformation was accomplished by extracting the 'male' column from the labels of the original dataset. The rationale for this adjustment stemmed from numerous challenges associated with handling a multi-label dataset, including increased processing requirements and computational complexities.

In addition, we have implement all functions of regularization methods used in our experiment. We build special data processor and dataloader for two datasets. We also construct our model structure specifying for Cifar10 and CelebA in order to have a better result.

The libraries we used are: Numpy, Matplotlib, Pandas, Pytorch, Random, and Math. We used an open source on github as our Resnet baseline. https://github.com/bearpaw/pytorch-classification.git

References

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